EXERCISE-2

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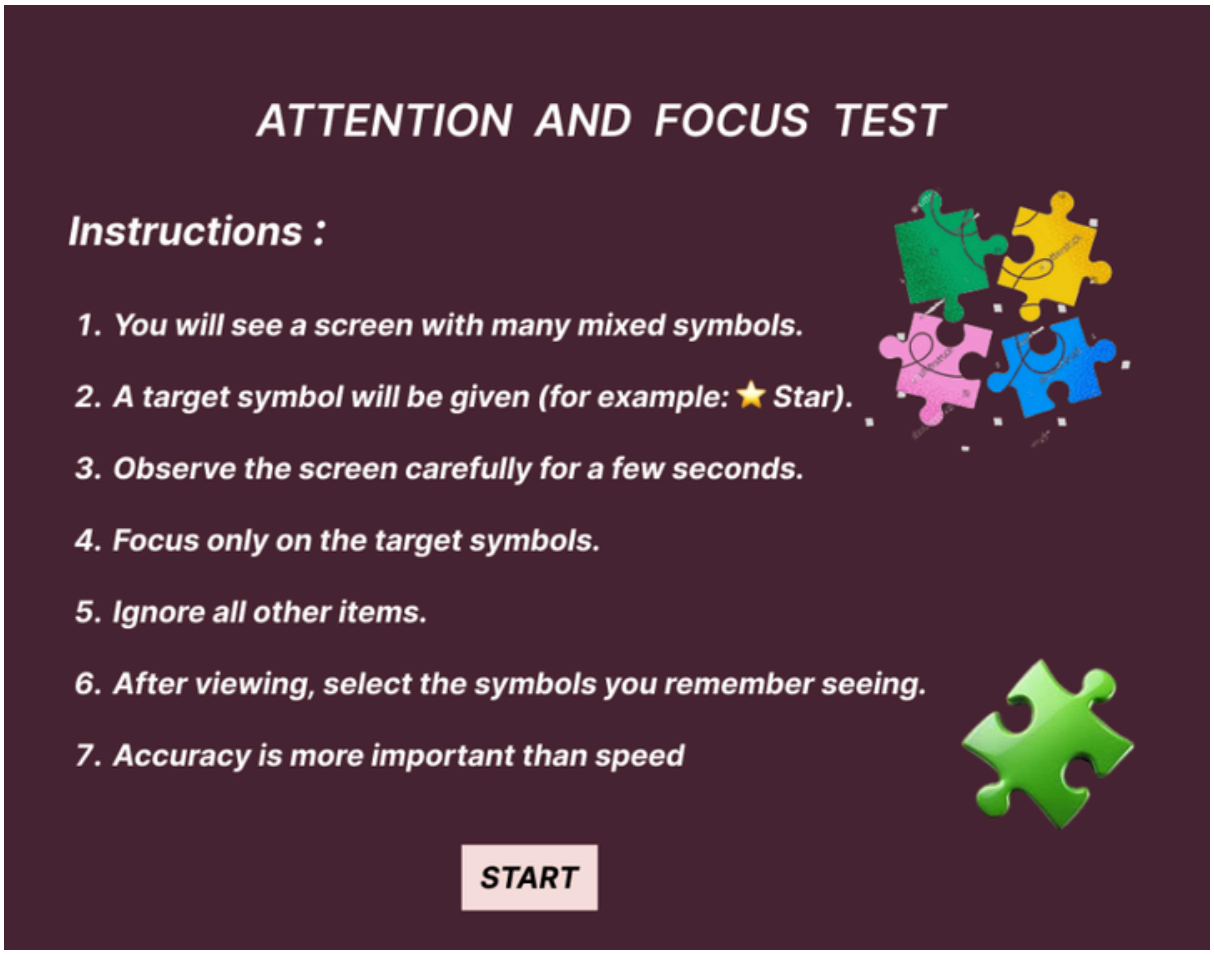
Design a UI where users recall visual elements (e.g., icons or text chunks). Evaluate the effect of chunking on user memory.

FRAME 1: INSTRUCTION PAGE

Analysis of the Attention and Focus Test – Instruction Screen

This screen represents the Instruction Phase of the Attention and Focus Test, where users are prepared for the task by clearly explaining what they need to observe and how they should respond. The instruction page applies principles of chunking and focused attention to reduce cognitive overload and improve task understanding.

Additionally, the presence of a START button clearly indicates the next action, making navigation intuitive. Decorative puzzle visuals subtly reinforce the idea of cognitive processing without distracting the user. Overall, this instruction screen effectively prepares users for the test by combining clarity, simplicity, and chunking principles to enhance attention and focus.



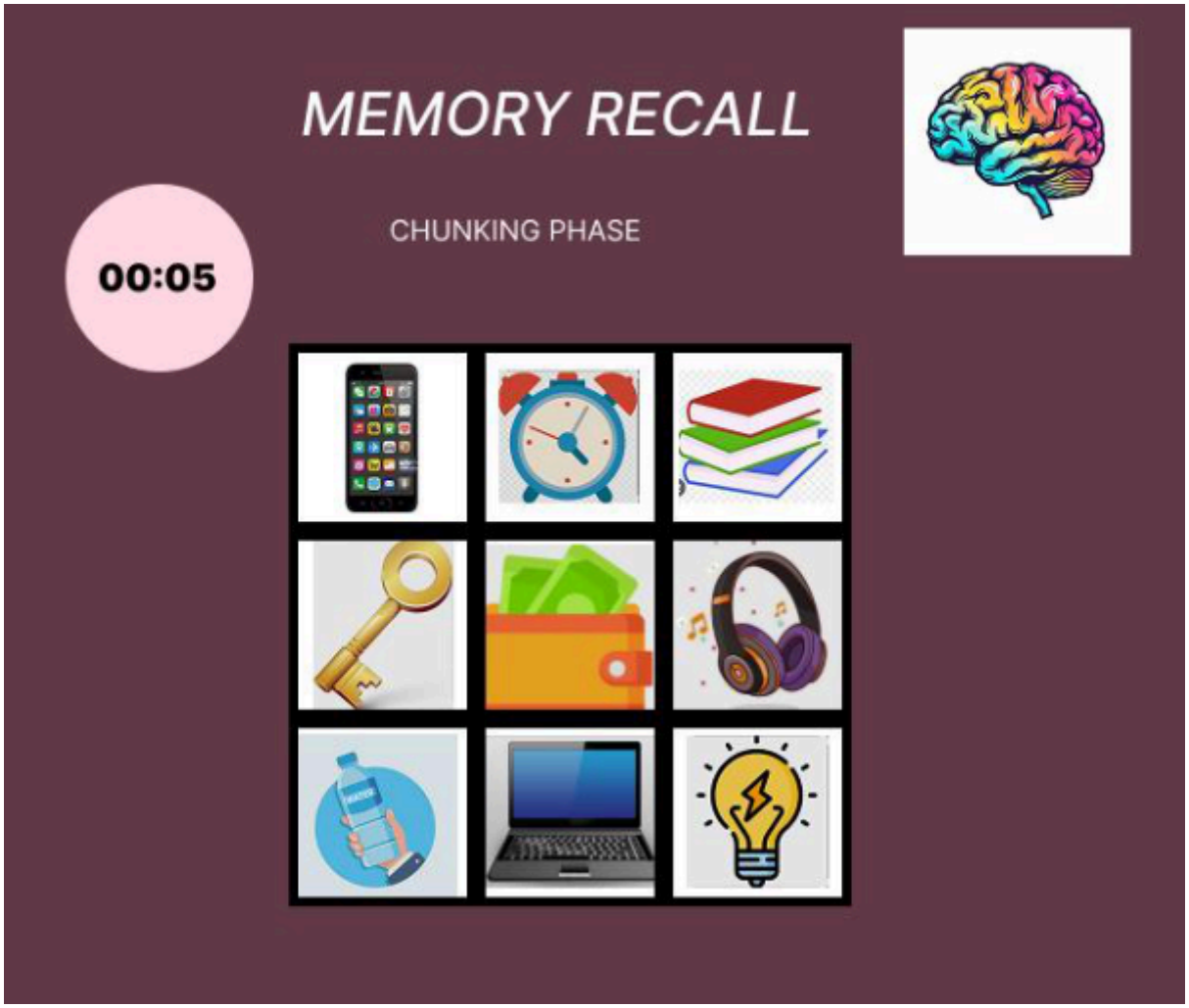
FRAME 2: CHUNKING PHASE

Analysis of the Memory Recall Task – Chunking Phase Screen

This screen represents the Chunking Phase of the Memory Recall Task, where users are briefly shown a structured set of visual items that they must observe and remember. The main objective of this phase is to enhance short-term memory by presenting information in organized chunks, rather than as random elements.

The screen displays a 3×3 grid of icons, including everyday objects such as a bottle, key, mobile phone, laptop, wallet, clock, bulb, headphones, and books. These familiar visuals reduce cognitive strain and allow users to associate items easily. Presenting the icons in a grid format supports the chunking principle, where related elements are grouped together to improve memory retention.

A countdown timer (00:05) is provided to limit observation time, encouraging focused attention and preventing over-analysis. This time constraint ensures that users rely on immediate visual memory, which is essential for testing recall ability. The clear labeling of “CHUNKING PHASE” reinforces the purpose of the screen and guides the user’s mental state toward observation rather than interaction.



Frame 3: Recall Phase – Selection Screen

Tests attention, focus, and short-term memory accuracy.

Key UI Elements

Title & Instruction

“ATTENTION AND FOCUS TEST”

“SELECT THE ITEMS YOU REMEMBER” – clearly guides user action.

Grid of Icon Choices

Multiple icons displayed in a structured grid.

Includes both previously shown items and distractor items.

Each icon has a selectable indicator (radio button/dot).

Submit Button

Allows users to confirm their selections and proceed.

Visual Design

Minimal, distraction-free layout to keep focus on recall.

How It Works

Users observe all displayed icons.

They identify and select the items they remember.

Distractors test false memory recognition.

Clicking SUBMIT completes the recall phase.

Cognitive & UX Benefits

Measures memory accuracy effectively.

Encourages focused thinking and decision-making.

Simple interaction reduces cognitive overload.

Gamified visuals keep users engaged.

Frame 4: Result Phase – Score & Feedback Screen

Overview

Frame 4 displays the results and feedback of the memory recall task. It helps users understand their performance and decide their next action.

Purpose of the Screen

Provides instant feedback on recall performance.

Displays the user’s recall score clearly.

Offers options to continue, retry, or exit the test.

Key UI Elements

Title & Score Display

“ATTENTION AND FOCUS TEST”

“YOUR RECALL SCORE” shown prominently.

Score displayed as 4/5, indicating correct selections.

Feedback Visual

Emoji/character gives positive reinforcement.

Action Buttons

Continue – move to the next stage.

Restart – retake the test to improve score.

Exit – end the task.

How This Phase Works

System evaluates user selections from Frame 3.

Calculates the recall score.

Displays performance feedback.

User chooses the next action based on results.

Cognitive & UX Benefits

Immediate feedback improves learning awareness.

Encourages self-assessment and improvement.

Clear navigation gives users control.

Positive visuals reduce stress and motivate users.